

Aalok Thakkar

Assistant Professor

DEPARTMENT OF COMPUTER SCIENCE

ASHOKA UNIVERSITY, SONIPAT, INDIA

thakkar@ashoka.edu.in • aalok-thakkar.github.io

Academic Appointments

2024–PRESENT	ASHOKA UNIVERSITY, Sonipat, India Department of Computer Science <i>Assistant Professor</i>
2023–2024	APTOS LABS AND MOVEMENT LABS, Seattle, WA, USA <i>Research Scientist, Programming Languages</i>
SUMMER 2021	AMAZON WEB SERVICES, Boston, MA, USA Automated Reasoning Group <i>Research Intern</i>

Education

2018–2023	UNIVERSITY OF PENNSYLVANIA Ph.D., Computer and Information Science Thesis: <i>Example-Guided Synthesis of Relational Queries</i> Advisors: Rajeev Alur and Mayur Naik
2015–2018	CHENNAI MATHEMATICAL INSTITUTE B.Sc. (Honours), Mathematics and Computer Science

Publications

Conference Publications

A Correctness-First Introduction to Computer Science

with Vaani Goenka

Innovation and Technology in Computer Science Education (ITiCSE 2026)

A Good Rubber Duck Does Not Quack

with Ayush Thonge

Innovation and Technology in Computer Science Education (ITiCSE 2026)

Accessibility Beyond Accommodations

with Vaanee Tripathi

ACM India Compute Conference (COMPUTE 2025)

Ancient Algorithms for Modern Curriculum

ACM India Compute Conference (COMPUTE 2025)

BOOP: Write Right Code

with Vaani Goenka

ACM India Compute Conference (COMPUTE 2025)

SocraticAI: Transforming LLMs into Guided CS Tutors Through Scaffolded Interaction

with Karthik Sunil

ACM India Compute Conference (COMPUTE 2025)

From Transformers to Weighted Automata: Towards Verification of Large Language Models

with Smayan Agarwal, Aslah Ahmad Faizi, and Shobhit Singh

From Data to Models and Back (DATAMOD 2025)

Mobius: Synthesizing Relational Queries with Recursive and Invented Predicates

with Nathaniel Sands, George Petrou, Rajeev Alur, Mayur Naik, and Mukund Raghothaman

Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA 2023)

Relational Query Synthesis $\bowtie$ Decision Tree Learning

with Aaditya Naik, Adam Stein, Rajeev Alur, and Mayur Naik

International Conference on Very Large Data Bases (VLDB 2023)

Example-Guided Synthesis of Relational Queries

with Aaditya Naik, Nathaniel Sands, Mukund Raghothaman, Mayur Naik, and Rajeev Alur

Programming Language Design and Implementation (PLDI 2021)

Additive Collatz Trajectories

with Mrunmay Jagadale

International Conference of the Jangjeon Mathematical Society (ICJMS 2016)

Journal Publications

Reopening businesses and risk of COVID-19 transmission

with Ashley O'Donoghue, Tenzin Dechen, Whitney Pavlova, Michael Boals, Garba Moussa, Manvi Madan, Frank J. DeFalco, and Jennifer P. Stevens

npj Digital Medicine 4, 51 (2021)

Concurrency in Boolean networks

with Thomas Chatain, Stefan Haar, Juraj Kolcak, and Loic Pauleve

Natural Computing 19, 91–109 (2020)

Infinitude of Primes Using Formal Languages

The American Mathematical Monthly 125:8, 745–749 (2018)

Workshop Publications

Complexity of Relational Query Synthesis

with Rajeev Alur and Mayur Naik

11th Workshop on Synthesis (SYNT 2022)

Modular Synthesis of Reactive Systems

with Kedar S. Namjoshi and Richard Trefler
9th Workshop on Synthesis (SYNT 2020)

Verification of Rewriting-based Query Optimizers
with Krishnamurthy Balaji and Piyush Gupta
16th International Workshop on Termination (WST 2018)

Pre-prints

When Does a Monoid Action Preserve Regularity?

Stochastic Languages at Sub-stochastic Cost
with Smayan Agarwal

Identity Testing for Stochastic Languages
with Smayan Agarwal and Shobhit Singh

Selected Talks

Proofs Without Proofs

07/2026 *ICHIM 2026, University of Peradeniya, Sri Lanka*

Formal Methods Update Meeting 2026

07/2026 *Krea University, Sri City, India (organizer and invited talk)*

Identity Testing for Stochastic Languages

2025 *IARCS Verification Seminar Series*

Synthesis of Relational Queries with Recursive and Invented Predicates

2023 *OOPSLA 2023, Lisbon, Portugal*

Relational Query Synthesis $\bowtie$ Decision Tree Learning

2023 *VLDB 2023, Vancouver, Canada*

Example-Guided Synthesis of Relational Queries

2021 *PLDI 2021, online*

Teaching

SUMMER 2026 *FC-0306: Quantitative Reasoning and Mathematical Thinking*

SPRING 2026	CS-3330: <i>Theory of Computation</i> CS-3610: <i>Information Security</i>
MONSOON 2025	CS-1102: <i>Introduction to Computer Science</i> CS-5310: <i>Symbolic Logic and Applications</i> CS-6101: <i>Computer Science Research Methods and Seminars</i>
SUMMER 2025	FC-0306: <i>Quantitative Reasoning and Mathematical Thinking</i> CS-1102: <i>Introduction to Computer Science</i>
SPRING 2025	CS-1102: <i>Introduction to Computer Science</i> CS-5440: <i>Trustworthy AI</i>
MONSOON 2024	CS-5310: <i>Symbolic Logic and Applications</i>

Grants and Fellowships

2026–	ANRF PRIME MINISTER EARLY CAREER RESEARCH GRANT <i>Robust and scalable program synthesis.</i>
2025–2026	DATA & AI ADOPTION FELLOWSHIP, EkStep Foundation <i>AI Safety Benchmarks for Education and Agriculture.</i>
2025–2026	DARJEELING REVIVAL FELLOWSHIP, Alt Carbon <i>Geostatistical modelling for enhanced rock weathering.</i>
2025–2026	ADAPTIVE AND INCLUSIVE PEDAGOGY GRANT, Mphasis <i>Curriculum design for an adaptive and inclusive classroom.</i>
2026	NAGARATHNA MEMORIAL GRANT <i>Translation of Nayi Kavita poets into English.</i>

Selected Opinion Pieces

Your OTP isn't a one-time password any more, and that carries a security cost

The Indian Express (2026)

Rejoinder: The retroflex, like the Marathi 'na', has little to do with caste

Scroll.in (2026)

New rules for living with intelligent machines

Hindustan Times (2026)

Computer science cannot rely on jugaad for real-world reliability

Moneycontrol (2025)

Staying relevant: A professor's insights on what makes engineers more employable

CNBC TV18 (2025)

Ensuring a Level Playing Field

Moneycontrol (2025)

Professional Service

ORGANIZER	<i>Formal Methods Update Meeting 2026, Krea University.</i> <i>FSTTCS 2024, FSTTCS 2025.</i> <i>IARCS Verification Seminar Series.</i> <i>Logic Society of Delhi.</i>
EDITOR	<i>Corresponding editor, Bhāvanā magazine.</i>
REVIEWER	<i>Programme committees and review activities for POPL AEC 2022, ISEC 2025, PLDI</i> <i>SRC 2025, ATVA AEC 2025, OOPSLA 2026, and ACM Computing Surveys.</i>
MEMBERSHIPS	<i>Association for Computing Machinery (ACM India).</i> <i>Vijnana Bharati.</i>